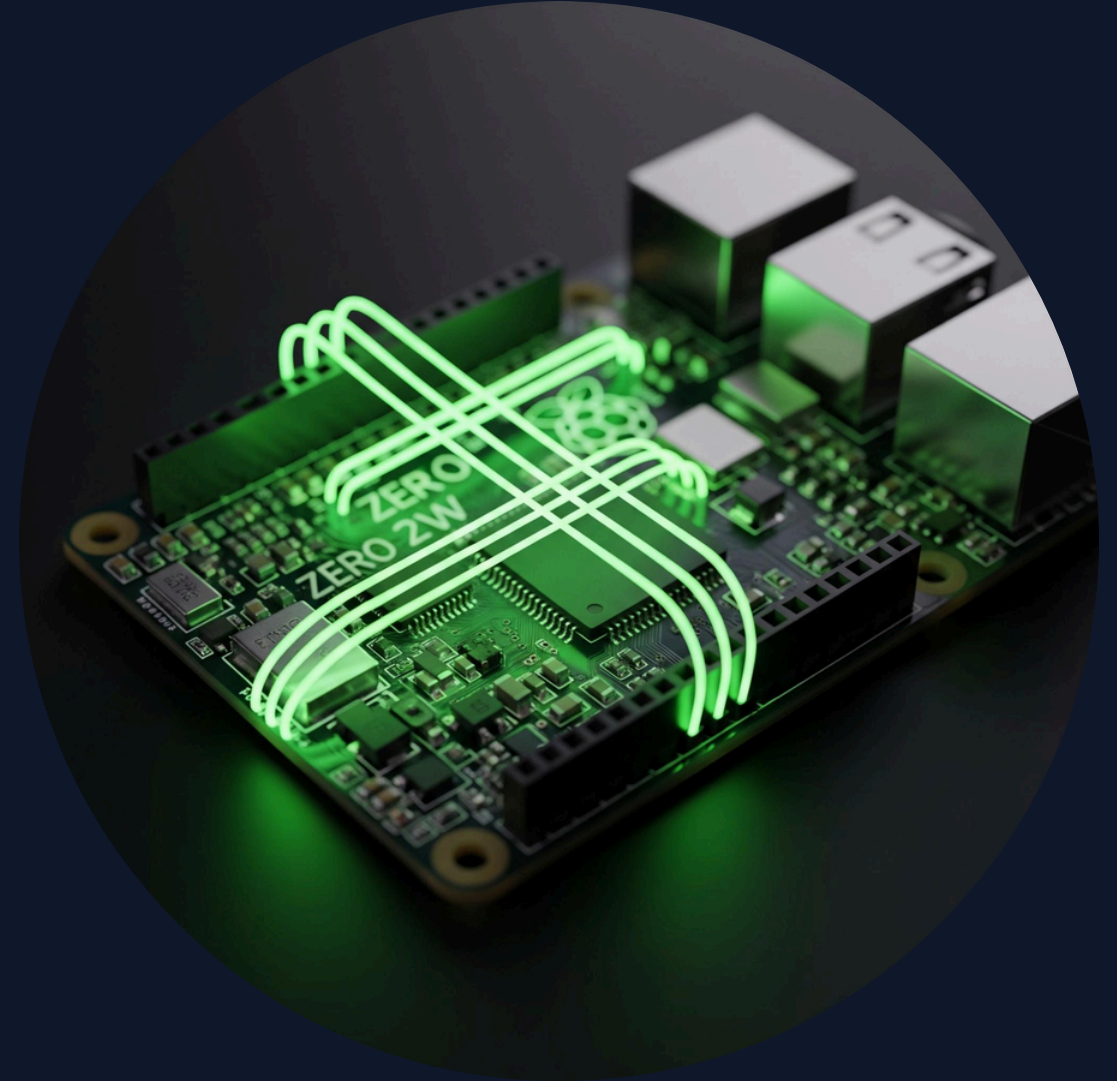


Fitness Progress Checker

Personalized Nutritional Intelligence via
Raspberry Pi Zero2W

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BEN 414857028



| What Does This Web App Do?



For Users:

1. Enter height & weight → Get BMI + calorie recommendation
2. Save records to database automatically
3. View all historical fitness logs
4. Automatically save records to database



For Admins:

- Install on RPi0 2W with DietPi
- Configure Nginx + PHP + MariaDB

3-Tier System Architecture



Presentation

Responsive UI built with Semantic HTML and Modern CSS. Facilitates user input and renders recommendation dashboards.



Application

PHP Server-side logic handling Mifflin-St Jeor calculations, data validation, and secure database transactions.



Data Layer

MariaDB relational storage utilizing normalized tables to maintain integrity across users, logs, and macros.

Database Design (3NF)



users:

user_id (PK)
name, age, gender



fitness_logs

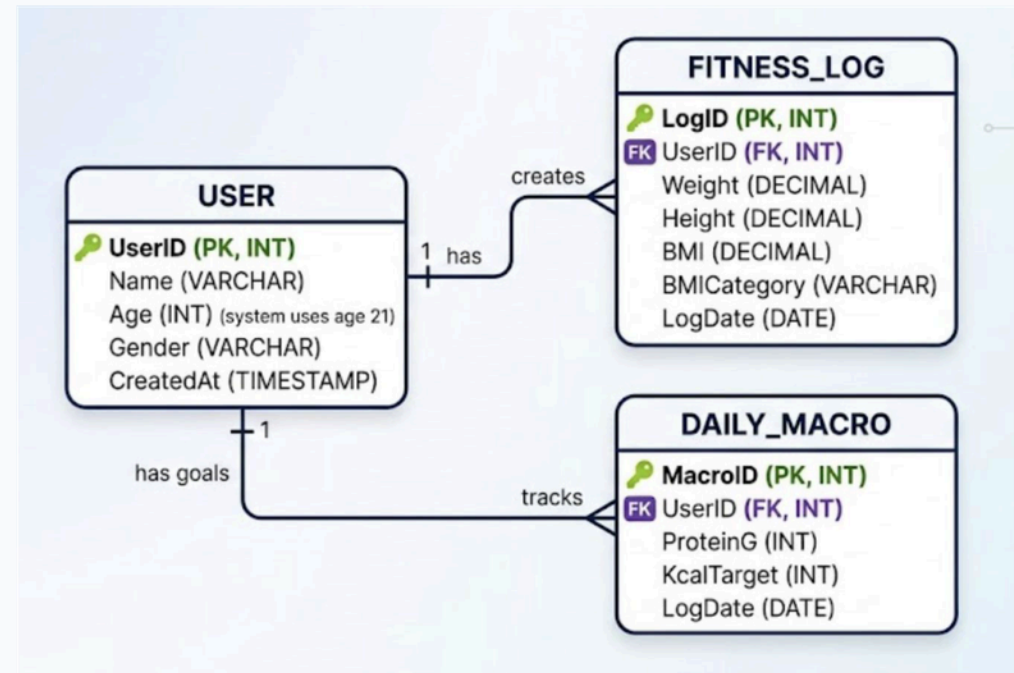
log_id (PK)
user_id (FK)
weight, height, BMI, category, date



daily_macros

macro_id (PK)
user_id (FK)
protein_g, kcal_target, log_date

Satisfies 3rd Normal Form



Hardware & Deployment



Hardware:

Raspberry Pi Zero 2W (512MB RAM)
Link-local network only (WiFi off)

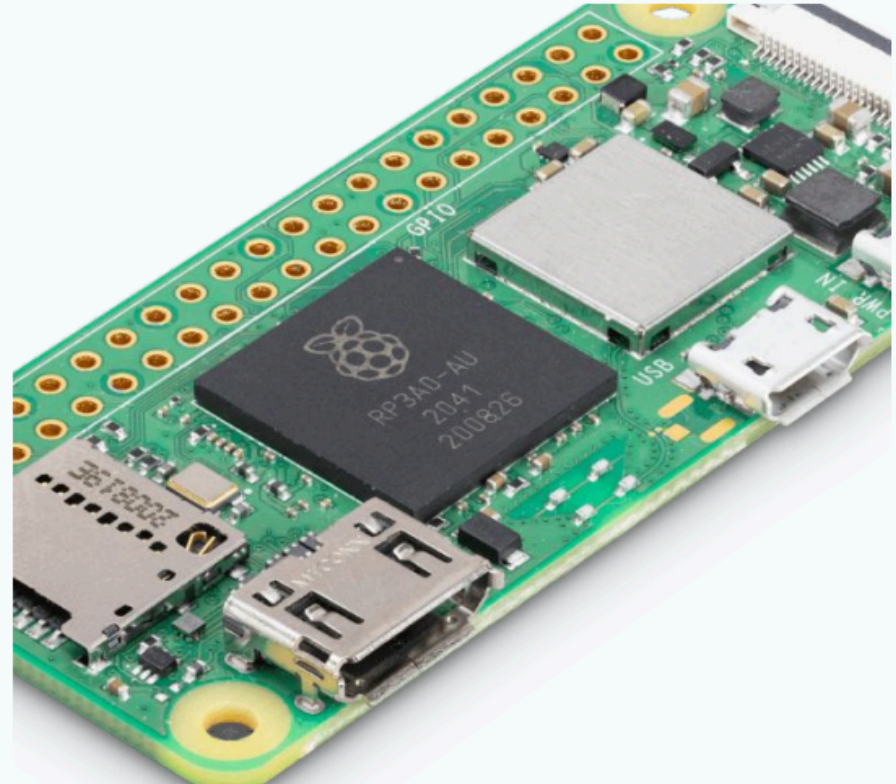


Software Stack:

DietPi OS
Nginx + PHP 7.4 + MariaDB

Quick Install:

1. git clone [repo URL]
2. Import schema.sql
3. Configure db.php



Logic: Mifflin–St Jeor Engine

BMR & TDEE Calculation:

$$\text{BMR} = (10 \times \text{weight}) + (6.25 \times \text{height}) - (5 \times \text{age}) + 5$$

Age constant: 21

$$\text{TDEE} = \text{BMR} \times \text{Activity Multiplier (1.55)}$$

Action Plan Driver

This data identifies if the user needs a 500 kcal deficit or a 300 kcal surplus for optimal health outcomes.

